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Etching could boost disk capacity

By Eric Smalley, Technology Research News

The laws of physics present challenges for making both ever-smaller circuits in computer chips and magnetic disk drives that can store more data.

To fit more information on a disk, researchers are looking for ways to make smaller magnetic domains, which are the portions of disks that hold individual bits of data. Most are using thin films of metal that contain clumps of tiny magnetic particles, with the clumps serving as the domains.

A team of researchers at the University of Cambridge in England is taking a different approach. They have found a way to make magnetic domains in homogenous thin films that contain no clumps. Homogenous thin films are easier and cheaper to produce than the lumpier sort used in today's magnetic media.

The key to making the thin film form the magnetic domains that represent the ones and zeros of computer data is using the interactions produced by layering different types of films. "In previous magnetic thin film research, the magnetic domains [were] defined by the magnetic properties of the material involved," said Anthony Bland, a professor of physics at the University of Cambridge. "We can control the local [magnetic] orientation in a chemically homogeneous film" using several different thin-film layers, he said.

The method could eventually increase the amount of data that can be crammed onto magnetic media by an order of magnitude.

To make the media, the researchers etched patterns of nickel oxide lines or dots into a gallium arsenide wafer using the lithography techniques used to make computer chips, then added a layer of copper cobalt followed by a layer of nickel. This layering changed the magnetic properties of the nickel.

At the microscopic level, the structure of the copper cobalt sitting above the nickel oxide patterns is jumbled, but the microstructure of the copper cobalt sitting directly on the wafer is orderly. The magnetic field of the thin film of nickel on top of the orderly copper cobalt is perpendicular but is parallel where the copper cobalt is jumbled.

The researchers used this construction to form separate domains. "The perpendicularly oriented domains can be used as storage bits... separated by a static magnetic wall from the in-plane magnetization," said Bland. The advantage of the researchers' approach is that the domains are formed not by creating a special media but by using existing chipmaking technology to form patterns below a simple magnetic film.

The researchers' prototype contained 7-micron magnetic domains spaced 3 microns apart, which is too large to make useful disk drives. Typical disk drives today use rectangular bits at 200- to 500-nanometer intervals. A nanometer is one millionth of a millimeter and one thousandth of a micron. Simulations show that the researchers' technique has the potential to produce square bits as small as thirty nanometers across, said Bland.

"Depending on the size of the lithographic pattern and the chosen magnetic materials, the bit resolution can be higher than 300 gigabits per square inch," said Bland. Three hundred gigabits is enough to hold about 18 hours of DVD-quality video.

In contrast, IBM's top-of-the-line disk technology holds about 25 gigabits per square inch. IBM researchers are aiming to push that to 100 gigabits per square inch using existing technology.

"It's a nice idea to use the microstructure of the film to alter the magnetic [orientation], and therefore to create regions where data can be stored," said Caroline Ross, an associate professor of materials science and engineering at the Massachusetts Institute of Technology.

"The main question is related to the size-scale achievable. For a patterned medium, you would need features of sub-50 nanometer dimensions," said Ross. "Microstructural control on such a small scale is difficult, and I can't tell at this stage whether it will work on a nanoscale. So I don't know whether this will ever be technologically useful in recording media," she said.

The scheme is compatible with current manufacturing processes and the magnetic medium is compatible with current magnetic data recording technology, said Bland. The technique could be incorporated into magnetic media technology within three years, he said.

Bland's research colleagues were Shunpu Li, Wen-Siang Lew, Luis Lopez-Diaz and Carlos Vaz of the University of Cambridge, and Marco Natali and Yong Chen of the National Center for Scientific Research (CNRS) in France. They published the research in the February 7, 2002 issue of the journal Nature. The research was funded by the European Community.

Timeline: 3 years

Funding: Government

TRN Categories: Data Storage Technology; Materials Science and Engineering

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